



CHALLENGE 02 / 20

# THE ERASED CLINICAL TRIAL

MANIPULATED AND DELETED RESEARCH RECORD RECONSTRUCTION

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

## 1 | PROBLEM BACKGROUND

A contract research organisation ran a Phase III cardiac trial across 62 sites in 11 states, enrolling 4,800 patients, for an Indian pharma company seeking marketing approval. A regulator's inspection found that outcome tables in the submission did not agree with site source records. Over 22 months the data passed through site entry, data-management queries, an external statistician, a cloud trial platform and three database migrations.

Records have been edited, merged, re-coded and deleted. The audit-trail tables were partly truncated in one migration. Backup rotation destroyed every copy older than nine months. Two staff hold administrator rights to the platform. Many changes were legitimate: query resolutions, protocol amendments and medical dictionary upgrades. Some may hide serious adverse events and deaths.

The company states that every change was legitimate. The regulator needs to know what the data said before it was changed, and whether the efficacy conclusion survives.

## 2 | PROBLEM STATEMENT

Develop a **Research Record Integrity Reconstruction System** that recovers the most probable original state of a regulated clinical dataset from degraded residual traces, and classifies every deviation as legitimate amendment, error or suspected tampering.

The system must also quantify how the trial's efficacy and safety conclusions would change if every unverifiable record were replaced by its most probable original, and under the worst case.

### Hard constraints

- The audit trail itself is partly destroyed, and for some records the original value exists nowhere.
- Legitimate dictionary re-coding and query resolutions must be told apart from concealment that imitates them.
- The reconstruction must work across thousands of patients and many sites, with three migrations in the history.
- Every recovered value must be traceable to its evidence, with stated uncertainty. No value may be presented as original without support.
- The conclusion-sensitivity analysis must be reproducible and must identify the smallest set of records whose history would change the result.

**No evidence package or dataset is provided.** Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

### 3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

|    |                                     |                                                                           |
|----|-------------------------------------|---------------------------------------------------------------------------|
| 01 | Record Version Reconstruction       | Ability to rebuild version histories from incomplete and migrated traces. |
| 02 | Original-Value Estimation           | Ability to estimate original values with explicit uncertainty.            |
| 03 | Change Intent Classification        | Ability to classify changes as amendment, error or concealment.           |
| 04 | Migration Artefact Handling         | Ability to tell migration side effects from human edits.                  |
| 05 | Safety-Signal Concealment Detection | Ability to detect changes that reduce visibility of adverse events.       |
| 06 | Audit-Trail Trust Assessment        | Ability to judge which parts of the audit trail can be trusted.           |
| 07 | Conclusion Sensitivity Analysis     | Ability to measure how conclusions depend on unverifiable records.        |
| 08 | Value-Level Provenance              | Ability to show how each recovered value was derived.                     |

The system must clearly distinguish between:

|          |           |         |                     |               |
|----------|-----------|---------|---------------------|---------------|
| VERIFIED | RECOVERED | AMENDED | SUSPECTED TAMPERING | UNRECOVERABLE |
|----------|-----------|---------|---------------------|---------------|

A recovered value that cannot be adequately supported must not be presented as the original.

**Build, don't search.** This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

## 4 | EXPECTED OUTPUT

**Live demonstration.** The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

|    |                                                                          |
|----|--------------------------------------------------------------------------|
| 01 | Teams' own simulated multi-site trial with ground-truth history          |
| 02 | Simulated edits, deletions, migrations and truncation of the audit trail |
| 03 | Recovery of original values with uncertainty                             |
| 04 | Classification of every deviation                                        |
| 05 | Separation of legitimate re-coding from concealment                      |
| 06 | Sensitivity of the efficacy and safety conclusions                       |
| 07 | The minimal set of records that would change the result                  |

The team must be able to explain why the system reached each major conclusion.

### Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

**Submission.** Each team must submit:

|    |                                      |                                                                                           |
|----|--------------------------------------|-------------------------------------------------------------------------------------------|
| 01 | <b>Working Prototype</b>             | A functional implementation of the proposed solution.                                     |
| 02 | <b>Source Code</b>                   | Complete source code required to reproduce the project.                                   |
| 03 | <b>Technical Architecture</b>        | Architecture and component description.                                                   |
| 04 | <b>Validation Environment</b>        | A controlled environment created by the team to demonstrate the solution.                 |
| 05 | <b>Technical Documentation</b>       | Implementation, design and testing documentation.                                         |
| 06 | <b>Forensic Investigation Report</b> | A professional report generated from the demonstrated investigation.                      |
| 07 | <b>Demonstration</b>                 | Live technical demonstration to the judging panel.                                        |
| 08 | <b>Limitations</b>                   | Known limitations and situations where the system cannot establish a reliable conclusion. |